Understanding Technical Indicators

Moving Averages: The Foundation of Technical Analysis

Moving averages are perhaps the most widely used technical indicators—think of them as the "ABCs" of technical analysis. They smooth out price fluctuations to help identify the direction of the trend while filtering out "noise" or short-term price fluctuations.

Imagine you're standing on a beach, watching waves crash against the shore. Each individual wave represents a daily price movement—sometimes higher, sometimes lower. But if you look at the overall waterline over time, you can see whether the tide is coming in (uptrend) or going out (downtrend). Moving averages help you see that "tide" by averaging out the individual waves.

There are several types of moving averages, each with different characteristics:

Simple Moving Average (SMA)

The Simple Moving Average is exactly what it sounds like—a simple average of prices over a specified period. For example, a 20-day SMA is the average closing price over the past 20 days. As each new day arrives, the oldest day drops off, and the newest day is added.

```
import pandas as pd
import numpy as np
import yfinance as yf
# Download historical data for Apple
apple = yf.download('AAPL', start='2022-01-01', end='2023-01-01')

# Calculate 20-day and 50-day Simple Moving Averages
apple['SMA20'] = apple['Close'].rolling(window=20).mean()
apple['SMA50'] = apple['Close'].rolling(window=50).mean()
```